CONTACT DETAILS OF THE BODY SUBMITTING THE QUALIFICATION FILE

Directorate General of Training (DGT)
Government of India, Ministry of Skill Development and Entrepreneurship,
1st and 2nd Floor, CIRTES Building
Next to Pusa ITI, Pusa Campus
New Delhi – 110012.

NCVET Code

2022/POW/DGT/06310

Name and address of submitting body:

Directorate General of Training (DGT)
Government of India, Ministry of Skill Development and Entrepreneurship,
1st and 2nd Floor, CIRTES Building
Next to Pusa ITI, Pusa Campus
New Delhi – 110012.

Name and contact details of individual dealing with the submission

Name: Mrs. Sandhya Salwan

Position in the organisation: Deputy Director General

Address if different from above:

Tel number(s): 011-25802140

E-mail address: sandhya.salwan@nic.in

List of documents submitted in support of the Qualifications File

1. Competency-based curriculum with following details:

Model Curriculum to be added which will include the following:
a) Indicative list of tools/equipment to conduct the training: Enclosed with curriculum
b) Trainers qualification: Indicated in the curriculum
c) Lesson Plan: All DGT curricula are designed indicating specific practical to be carried out during training along with details of trade theory. Based on this the concerned instructor prepares the Lesson Plan and demonstration plan with support of IMPs developed by NIMI,DGT.
d) Distribution of training duration into theory/practical/OJT component: Indicated in the curriculum.

2. Curriculum for Core Skills (Employability Skills).

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- **SUMMARY**

1	**Qualification Title**	**'WIREMAN'**
2	**Qualification Code, if any**	**DGT/1009**
3	**NCO code and occupation**	7411.0301 – Wireman, Light and Power
4	**Nature and purpose of the qualification (Please specify whether qualification is short term or long term)**	Prepare skilled Technician to undertake the job roles of Wireman and will enable the trainee to plan, install and repair electrical wiring such as cleat, conduit, casing, concealed etc in factories, workshops, business and residential premises etc. It is a long term qualification.
5	**Body/bodies which will award the qualification**	Directorate General of Training (DGT).
6	**Body which will accredit providers to offer courses leading to the qualification**	Directorate General of Training (DGT) accredits the Training providers (ITIs/ NSTIs/ MSTIs/ BTCs/ BTPs / Industries / Establishments).
7	**Whether accreditation/affiliation norms are already in place or not, if applicable (if yes, attach a copy)**	Yes. The accreditation/ affiliation norms and any amendments made from time to time are available on DGT web portal.
8	**Occupation(s) to which the qualification gives access**	● Wireman, Light and Power
9	**Job description of the occupation**	Wireman will be able to plan, install and repair electrical wiring, erect switch boards and fix switch box casings cleats, conduits, meters etc. in factories, workshops, business and residential premises etc. and the technician generally fixes fuses, covers as per diagram and eliminates chances of short circuit and earthing.
10	**Licensing requirements**	NOT REQUIRED
11	**Statutory and Regulatory requirement of the relevant sector (documentary evidence to be provided)**	NOT APPLICABLE
12	**Level of the qualification in the NSQF**	Level 4

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<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>13</td><td>Anticipated volume of training/learning required to complete the qualification</td><td>
<table>
<tr><th rowspan="2">S No.</th><th rowspan="2">Course Element</th><th colspan="2">Notional Training Hours</th></tr>
<tr><th>1st Year</th><th>2nd Year</th></tr>
<tr><td>1</td><td>Professional Skill (Trade Practical)</td><td>840</td><td>840</td></tr>
<tr><td>2</td><td>Professional Knowledge (Trade Theory)</td><td>240</td><td>300</td></tr>
<tr><td>3</td><td>Employability Skills</td><td>120</td><td>60</td></tr>
<tr><td></td><td>Total</td><td>1200</td><td>1200</td></tr>
</table>
<table>
<tr><td>On the Job Training (OJT)/ Group Project</td><td>150</td><td>150</td></tr>
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<tr><td>14</td><td>Indicative list of training tools required to deliver this qualification</td><td>As per Annexure-I of curriculum.</td></tr>
<tr><td>15</td><td>Entry requirements and/or recommendations and minimum age</td><td>Passed 8th Class.

Minimum age 14 years as on first day of academic session.</td></tr>
<tr><td>16</td><td>Progression from the qualification (Please show Professional and academic progression)</td><td>An Individual can proceed for:
<table>
<tr><td>Professional
● Wireman
● Senior Wireman
● Supervisor
● Manager
● Entrepreneur</td><td>Technical / Academic
|
|
ATS
CITS
10th/ Diploma/ Advance Diploma (Vocational)</td></tr>
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17	**Arrangements for the Recognition of Prior learning (RPL)**	● Yes (For more details refer "Guidelines for Private candidate" in DGT website MIS portal).
18	**International comparability where known (research evidence to be provided)**	-----
19	**Date of planned review of the qualification.**	3 Yrs from the Date of Approval
20	**Formal structure of the qualification** **Mandatory components**	

	Title of component and identification code/NOSs/ Learning Outcomes	**Estimated size (learning hours)**		**Level**
		Skills	**Knowledge**	
	TRADE SPECIFIC			
(i)	Apply safety precautions and prepare profile with an appropriate accuracy as per drawing using basic jobs of marking components, filing, drilling, riveting, fitting, joining etc. (NOS: PSS/N1707)	110	20	4
(ii)	Prepare terminations, make good quality of electrical wire joints for single and multi strand conductors and carry out crimping, soldering and brazing. (NOS: PSS/N2512,PSS/N1331)	60	10	4
(iii)	Draw and set up DC and AC circuits, involving R-L-C components, perform measurement of various electrical parameters with due care and safety. Carry out Sealing of energy meters and Monitor meter readings using MRI. (NOS:PSS/N1707)	130	30	4
(iv)	Explain basic concepts of generation, transmission and distribution of electrical power including renewable energy. (NOS: PSS/N7001)	50	10	4

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(v)	Plan and prepare Plate and Pipe earthing installations and ensure safe and effective earthing. (NOS: PSS/N6002)	40	7	4
(vi)	Carry out wiring, testing, and maintenance of DC machines including DC motor starters. (NOS: PSS/N9411)	50	10	4
(vii)	Carry out wiring, testing, and maintenance of small transformers, 1φ& 3φ AC motors and Alternators including AC motor starters. (NOS: PSS/N9412)	60	10	4
(viii)	Read, understand and draw electrical Schematic drawings of power and control circuits using industry standard symbols. (NOS: PSS/N9413)	50	10	4
(ix)	Plan, draw, assemble and perform various domestic wiring. Carry out Testing, maintenance and repair/ replacement of domestic wiring. (NOS: PSS/N9414)	175	35	4
(x)	Carry out wiring of control panels, assemble accessories and equipment. (NOS: PSS/N1709)	80	18	4
(xi)	Install, test and carry out maintenance of batteries and solar cell with due care and safety. (NOS: PSS/N6003)	35	10	4
(xii)	Read and apply engineering drawing for different application in the field of work. (NOS: PSS/N9401)	-	40	4
(xiii)	Demonstrate basic mathematical concept and principles to perform practical operations. Understand and explain basic science in the field of study. (NOS: PSS/N9402)	-	30	4
(xiv)	Plan, draw, install and test different types of Commercial wiring including advanced systems. Install temporary electrical wiring at construction site. (NOS: PSS/N1707)	115	30	4
(xv)	Plan, draw, estimate material/ cost, install and test different types of industrial wiring system as per IE rules. Layout cables for various purposes including cable management. (NOS: PSS/N1707)	110	28	4

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(xvi)	Plan, install and test illumination system including domestic, commercial and industrial requirements. Connect, program and operate PAR light on DMX controller (Stage light control). (NOS: PSS/N1707)	60	20	4
(xvii)	Assemble simple electronic circuits, repair CFL, LED lamps and DC regulated power supply. (NOS: PSS/N6002)	65	20	4
(xviii)	Assist in Installation and commissioning of small solar plant, solar pumps and construct Solar DC appliances. (NOS: PSS/N6003)	80	20	4
(xix)	Plan, prepare and carry out jointing of LT/HT underground cables with due care and safety. (NOS: PSS/N2512)	85	20	4
(xx)	Install Electric Vehicle charging stations and carry out preventive/breakdown maintenance. (NOS: PSS/N9410)	20	5	4
(xxi)	Install and repair domestic appliances viz., electric kettle, food processor, fan, washing machine, geyser, water pump etc. including repair of electrical faults in refrigerator, window and split AC. (NOS: PSS/N6003, PSS/N4402, PSS/N1711)	135	40	4
(xxii)	Perform winding of small transformers and motors viz., ceiling fan, table fan, mixer/grinder, submersible pump, etc. (NOS:PSS/N4402)	130	35	4
(xxiii)	Carry out Estimation & costing for different wiring systems and ready to adopt structured / smart wiring concept for automation and IoT applications. (NOS: PSS/N9413)	40	10	4
(xxiv)	Read and apply engineering drawing for different application in the field of work. (NOS: PSS/N9401)	-	40	4
(xxv)	Demonstrate basic mathematical concept and principles to perform practical operations. Understand and explain basic science in the field of study. (NOS: PSS/N9402)	-	32	4
CORE SKILL				

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EMPLOYABILITY SKILLS				
(i)	Introduction to Employability Skills	-	180	-
(ii)	Constitutional values - Citizenship	-		-
(iii)	Becoming a Professional in the 21st Century	-		-
(iv)	Basic English Skills	-		-
(v)	Career Development & Goal Setting	-		-
(vi)	Communication Skills	-		-
(vii)	Diversity & Inclusion	-		-
(viii)	Financial and Legal Literacy	-		-
(ix)	Essential Digital Skills	-		-
(x)	Entrepreneurship	-		-
(xi)	Customer Service	-		-
(xii)	Getting Ready for Apprenticeship & Jobs	-		
		2400		
	On the Job Training (OJT)/ Group Project	**300**		

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SECTION 1
ASSESSMENT

21	**Body/Bodies which will carry out assessment:** Controller of Examinations, DGT
22	**How will RPL assessment be managed and who will carry it out?** DGT will carry out the RPL assessment following the below mentioned eligibility criteria for Trainee: Applicants aspiring to appear as Private Candidates in the AITT under CTS for award of NTC, have been categorized based on their educational background and experience. Subsequently 'Private Candidates' may be admitted under one of the following categories. Category wise 'eligibility criteria' for appearing as 'Private Candidate' in AITT under CTS has been listed below: Category I: Ex-trainees (successful pass-outs) of ITI A. Ex-trainees of ITI who already possess NTC in one of the trades under CTS, are eligible for applying as Private candidate for an allied trade, provided he/ she fulfils all the conditions regarding educational qualification etc. prescribed for that allied trade. B. In addition, the applicant should possess minimum of 1 year experience (as on date of submission of application) post the date of AITT result declaration in the desired allied trade in establishments implementing Apprenticeship Training Scheme (ATS)/ establishments registered under the Apprenticeship portal or registered MSMEs or Entities registered with any government/local authorities / shops covered under Factories Act 1948 and Shops and Establishments Act applicable for the concerned State. Category II: 'Ex-trainees (successful pass-outs) and current trainees under CoE scheme A. The applicant should have the minimum prescribed entry qualification and should fulfil eligibility criteria for the desired trade under CTS, in which he/she intends to appear for AITT as Private Candidate. CoE candidates must register as 'Private Candidate' under CTS in the relevant/mapped CTS trade only. B. There should be a minimum gap of 1 year between successful completions of CoE training i.e. from the date of result declaration to the date of submission of application for 'Private Candidate' certification. C. During this gap of 1 year, the candidate must have undergone Industry training or gained experience in desired trade in establishments implementing Apprenticeship Training Scheme (ATS)/ establishments registered under the Apprenticeship portal or registered MSMEs or Entities

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	registered with any government/local authorities / shops covered under Factories Act 1948 and Shops and Establishments Act applicable for the concerned State. Category III: SCVT Candidates (admitted till August 2018 session) A. No special provisions have been made for SCVT Trainees to enrol as 'Private Candidate'. Going forward, SCVT trainees have been granted equivalence vide G.S.R 186(E) dated 2nd March 2017 for undergoing apprenticeship training under the Apprentices Act 1961 to obtain 'NAC'. B. Only for SCVT trainees admitted till August 2018 batch, provision has been made for obtaining NTC by appearing in AITT under 'Private Candidate'. Such trainees will continue to be governed by old guidelines for 'Private Candidate'. Category IV: Other Candidates (candidate not falling in any of the above 3 categories, including SCVT trainees enrolled from admission session 2019 onwards) A. The applicant should have the minimum prescribed entry qualification and should fulfil eligibility criteria for the relevant trade under CTS, in which he/she desires to appear for AITT as Private Candidate. B. Applicant should be minimum 21 years of age on the date of submission of application. There is no upper age limit. C. The applicant should possess minimum of 3 years' experience (on the date of submission of application) in the relevant trade in establishments implementing Apprenticeship Training Scheme (ATS)/ establishments registered under the Apprenticeship portal or registered MSMEs or Entities registered with any government/local authorities / shops covered under Factories Act 1948 and Shops and Establishments Act applicable for the concerned State. (For detail and updated information please refer to DGT web portal.)
23	**Describe the overall assessment strategy and specific arrangements which have been put in place to ensure that assessment is always valid, reliable and fair and show that these are in line with the requirements of the NSQF.** (1) **Assessment process:** The assessment for the qualification is carried out by conducting formative assessments, and end of year examinations (Summative). The formative assessments in respect of each Learning Outcome for practical and related theory are conducted by the concerned instructors for evaluating the knowledge and skill acquired by trainees and the behavioural transformation

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of the trainees. This formative assessment is primarily carried out by collecting evidence of competence gained by the trainees by evaluating them at work based on assessment criteria, asking questions and initiating formative discussions to assess understanding and by evaluating records and reports. Summative assessment is carried out by All India Trade Test on Trade Theory, Trade practical and Employability Skills. The question papers for the theory Examinations contain objective type questions.

The marking pattern and distribution of marks for the qualification are as under:

Marking Pattern				
Sl. No.	Type of assessment	Subject for the trade test	Marks for the 1st Year	Marks for the 2nd Year
1	Summative Assessment	Practical	250	250
2		Trade Theory	100	100
3		Employability Skills	50	50
4	Formative assessment based on Learning Outcomes		200	200
		TOTAL:	600	600

(2) **Minimum pass marks:**

The minimum pass percent for Trade Practical and Formative assessment is 60% & for all other subjects is33%. There will be no Grace marks.

Testing and certifications for the course:
Controller of examinations, DGT carries out the assessment and issues National Trade Certificate (NTC) following the norms and guidelines issued by the Directorate from time to time.

Overall assessment strategy:

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	Assessment of the qualification evaluates trainees to show that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The trainees may choose the preferred language for assessment. The underlying principle of assessment is fairness and transparency. While assessing the trainee, assessor is directed to assess as per the defined assessment criteria against the learning outcomes. The evidence of the competence acquired by the trainees can be obtained by conducting theory and practical examinations, observing the trainees at work, asking questions and initiating discussions to assess, understand and evaluate records and reports. The ultimate objective of the assessment is to assess the candidates as per the defined assessment criteria for the learning outcomes. **Specific Arrangements for assessment:** • Assessment is outcome-based. • There are formative and summative assessments in Theory and Practical. • Assessment is carried out in Trade theory, Trade Practical, Workshop Calculation and Science, Engineering Drawing and Employability Skills. • While Trade Theory and Trade Practical are used for assessing Trade-related jobs, Workshop Calculation and Science is used to test trainee's numerical and logical skills, Drawing is used to test the ability of the trainee to draw and read sketches and Employability skills is used to test the communication, professional language, leadership, entrepreneurship and team-work abilities of the trainee. • In addition to demonstration of theory and practical knowledge, trainees get a chance to present total personality. **Quality assurance activities:** Question papers are set by external paper setters/ software generated. Evaluation of Theory Examinations in Trade and Employability Skill is done by third-party agency. Trade Practical is examined by External Examiner.

24. Assessment evidences

Title of Component: Formative Assessment Breakup
(on half yearly average of the learning assessment covered)

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Means of assessment

Assessment will be evidence based comprising the following for each Learning Outcome:

Serial No.	Terminal Competency	Maximum Weightage (%)
1	Safety consciousness	15
2	Workplace hygiene	5
3	Attendance/ Punctuality	10
4	Ability to follow Manuals/ Written instructions	5
5	Application of Knowledge	10
6	Skills to handle tools / equipment/ Instruments/ Devices	10
7	Economical use of materials	5
8	Working Strategy	10
9	Quality in workmanship/ Performance	15
10	VIVA	15
	Total Maximum Weightage (%)	100

Pass/Fail

The minimum pass percentage is 60% marks for formative assessment.

LEARNING OUTCOME WITH ASSESSMENT CRITERIA:

LEARNING OUTCOME (TRADE SPECIFIC)	
LEARNING OUTCOMES	**ASSESSMENT CRITERIA**
First Year	
1. Apply safety precautions and prepare profile with an appropriate accuracy as per drawing using basic jobs of marking components, filing, drilling, riveting, fitting, joining etc. (NOS: PSS/N1707)	Identify trade tools and equipment; demonstrate their uses with safety, care & maintenance.
	Identify safety symbols and hazards.
	Procedure of fire fighting in case of electrical fire.
	Make a wooden switchboard.
	Prepare a closed cabinet from metal sheet with holes for cables and various fittings.
2. Prepare terminations, make good quality of electrical	Identify types of wires, cables and their specifications.
	Measure size of the wire using SWG /micrometer.

wire joints for single and multi strand conductors and carry out crimping, soldering and brazing. (NOS: PSS/N2512, PSS/N1331)	Make married and 'T' (Tee) joint in stranded conductors.
	Prepare a Britannia straight and 'T' (Tee) joint in bare conductors.
	Prepare western union joint in bare conductor.
	Prepare Rat tail/ Duplex cross/ Knotted type/ fixture Joints in bare conductor.
	Solder the finished copper conductor joints with precaution.
	Prepare termination of cable lugs by using crimping tool.
	Demonstrate joining of metals by brazing.
3. Draw and set up DC and AC circuits, involving R-L-C components, perform measurement of various electrical parameters with due care and safety. Carry out Sealing of energy meters and Monitor meter readings using MRI. (NOS:PSS/N1707)	Measure resistance using voltage drop/Wheatstone bridge method.
	Measure current and voltage in electrical circuits and verify Kirchhoff's Law.
	Verify the characteristics of series-parallel combination of resistors.
	Wind a solenoid, determine the poles and plot the field of a magnet bar.
	Demonstrate generation of mutually induced emf.
	Measure current, voltage, power factor and determine the characteristics of RL/ RC / RLC in AC series / parallel circuits.
	Measure power, energy for lagging / leading power factors in single phase / three phase circuits.
	Demonstrate improvement of PF by use of capacitors in AC three phase circuits.
	Find the phase sequence of 3-phase supply using phase sequence meter.
	Measure the Power of three phase circuit for balanced and unbalanced loads
	Measure Power/ Energy/ Frequency/Current using Wattmeter/ Energy meter / Frequency/ Tong tester meter in single and three phase circuits.
	Use analog /digital multi-meter for measurement of different electrical parameters.
	Explain installation and sealing of energy meters and readings using MRI.
4. Explain basic concepts of generation, transmission	Make a block diagram of Thermal /Solar/ wind/ small, mini & micro hydro power plants/ Nuclear power plants.

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and distribution of electrical power including renewable energy. (NOS: PSS/N7001)	Make line diagram of transmission and distribution systems.
	Identify major equipment used in different substations viz., outdoor, indoor, pole mounted, etc.
	Prepare a line diagram of the institute/ ITI supply system.
5. Plan and prepare Plate and Pipe earthing installations and ensure safe and effective earthing. (NOS: PSS/N6002)	Identify various components of different earthing system.
	Measure earth resistance by earth tester/ megger.
	Perform grounding of equipment and systems.
	Test earth leakage by ELCB and relay.
6. Carry out wiring, testing, and maintenance of DC machines including DC motor starters. (NOS: N/A)	Identify parts of DC machines/ DC motor starters and their terminals.
	Carry out wiring of given DC motor / generator.
	Explain Service and repair of three point / four-point DC motor starters.
	Perform maintenance of carbon brushes, brush holders, Commutator and slip-rings.
	Perform speed control of DC motors - field / armature control method.
	Demonstrate overhauling/ routine maintenance of DC machines.
7. Carry out wiring, testing, and maintenance of small transformers, 1φ& 3φ AC motors and Alternators including AC motor starters. (NOS: N/A)	Identify terminals, components of single phase / three phase transformers and carry out wiring.
	Carry out polarity/ insulation/ open circuit/ short circuit test /voltage regulation of a transformer.
	Identify parts and terminals of single phase / three phase AC motors, test for continuity / insulation resistance.
	Identify parts and terminals of MG set and make connections.
	Identify parts and service of AC motor starters DOL/ star-delta/ auto-transformer /rotor resistance starter.
8. Read, understand and draw electrical Schematic drawings of power and control circuits using industry standard symbols. (NOS: N/A)	Draw symbols used in the electrical circuit drawings.
	Interpret control and power circuits of given wiring drawings.
	Draw circuit for control of lamps/ tube lights/ fans / single phase motors.
	Draw a circuit of fully automatic star-delta starter for starting a 3-φ induction motor.

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9. Plan, draw, assemble and perform various domestic wiring. Carry out Testing, maintenance and repair/ replacement of domestic wiring. (NOS: N/A)	Calculate maximum connected load in a section of the institute.
	Draw electrical supply system from pole to main switch board.
	Wire up PVC Casing-capping wiring to control one lamp from two different places (Staircase wiring).
	Wire up PVC conduit wiring to control one lamp from three different places.
	Prepare main distribution board, mount the energy meter board.
	Wire up the consumers main board with ICDP switch and distribution fuse box.
	Carry out earth continuity test.
	Check line-earth and neutral-earth loop impedance.
	Tracing of simulated faults in given circuit.
10. Carry out wiring of control panels, assemble accessories and equipment. (NOS: N/A)	Carry out wiring of Electrical panel, mount various control elements and secure the cables properly.
	Explain electro-magnetic interference and electro-magnetic compatibility.
	Perform wiring of control panel for different operations/controls of motor using various accessories and test for its performance.
11. Install, test and carry out maintenance of batteries and solar cell with due care and safety. (NOS: PSS/N6003)	Carry out charging of a Lead acid cell/ filling of electrolytes, testing of charging/ checking of discharged and fully charged battery.
	Explain routine, care/ maintenance and testing of batteries.
	Identify different types of solar cell viz., a-Si, CdTe, c-Si, CI(G)S, CVP and HCVP, etc.
	Determine the number of solar cells in series/ parallel for given power requirement.
12. Read and apply engineering drawing for different application in the field of work.	Read & interpret the information on drawings and apply in executing practical work.
	Read &analyze the specification to ascertain the material requirement, tools and assembly/maintenance parameters.

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	Encounter drawings with missing/unspecified key information and make own calculations to fill in missing dimension/parameters to carry out the work.
13. Demonstrate basic mathematical concept and principles to perform practical operations. Understand and explain basic science in the field of study.	Solve different mathematical problems
	Explain concept of basic science related to the field of study
SECOND YEAR	
14. Plan, draw, install and test different types of Commercial wiring including advanced systems. Install temporary electrical wiring at construction site. (NOS: PSS/N1707)	Carry out wiring for communication circuits and computer networks using UTP, STP, Co-axial and optical fibre cables.
	Wire-up lighting system for control using motion detector.
	Wire-up panel board for control of lights and fans from wireless remote.
	Install 1 φ / 3 φ online/ offline UPS wiring and test.
	Install and wire up CCTV camera.
	Install inverter and carry out wiring.
	Explain wiring planfor bathing area.
	Explain multi-storeyed building wiring.
	Install temporary LV electrical panels and lighting arrangements for construction site.
15. Plan, draw, estimate material/ cost, install and test different types of industrial wiring system as per IE rules. Layout cables for various purposes including cable management. (NOS: PSS/N1707)	Identify accessories and tools required for industrial wiring.
	Determine minimum ampacity and size of conductors for continuous and non-continuous loads.
	Demonstrate cutting, threading and bending of metallic conduit.
	Identify different bus bars, joining and installation including overhead bus bar system as per IE rules.
	Prepare bill of material, plan and practice wiring of an institute and workshop as per IE rules.
	Demonstrate split cable entry for multiple pre-terminated cables, up to IP 65 rated protection.
	Perform bonding and grounding of raceways, cable assembly and panels.

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	Demonstrate use of earth rods. Explain testing of underground cables for faults and removing of the fault.
16. Plan, install and test illumination system including domestic, commercial and industrial requirements. Connect, program and operate PAR light on DMX controller (Stage light control). (NOS: PSS/N1707)	Install fluorescent tube/ LED linear display lighting/ LED exhibition lighting / Pendant lighting.
	Prepare decorative lamp circuit to produce rotating/ running light effect.
	Install display spotlights and LED downlights.
	Explain/Demonstrate kitchen under-cabinet lighting, shelf lighting, closet lighting and cove lighting.
	Install lamps; HP mercury vapour / LP mercury vapour/ HP sodium vapour/ LP sodium vapour/ metal halide.
	Assemble and program DMX controller for operation of PAR lights.
17. Assemble simple electronic circuits, repair CFL, LED lamps and DC regulated power supply. (NOS: PSS/N6002)	Determine the value of resistance by colour code and identify types.
	Determine V-I characteristics of semiconductor diode.
	Identify circuit components and their terminals viz, diode, transistor, capacitors, regulator etc.
	Construct half wave/ full wave / bridge rectifier.
	Troubleshoot defects in simple power supplies.
	Identify different components and explain circuits of CFL & LED lamps.
	Perform repairing of LED / CFL.
18. Assist in Installation and commissioning of small solar plant, solar pumps and construct Solar DC appliances. (NOS: PSS/N6003)	Construct a solar lantern using Solar PV panel.
	Construct a Solar Day lighting using manual charge controller.
	Construct a Solar Street light using dusk to dawn charge controller.
	Construct a Solar water pump.
	Connect a Solar panel, Solar charge controller, Solar battery and a normal inverter and convert to a solar inverter.
	Prepare bill of material for a 1 KW solar PV installation.
	Explain synchronization between Solar Panel & AC grid supply.
19. Plan, prepare and carry out jointing of LT/HT	Identify different parts of various underground cables.
	Prepare cable for termination and joining.

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underground cables with due care and safety. (NOS: PSS/N2512)	Explain discharging procedure of underground cables.
	Make straight joint of underground cable.
	Explain testing of underground cables.
20. Install Electric Vehicle charging stations and carry out preventive/breakdown maintenance. (NOS: N/A)	Explain charger specifications.
	Install EV charging Station for public place.
	Install EV charging Station for home.
21. Install and repair domestic appliances viz., electric kettle, food processor, fan, washing machine, geyser, water pump etc. including repair of electrical faults in refrigerator, window and split AC. (NOS: PSS/N6003, PSS/N4402, PSS/N1711)	Service and repair of bell/ buzzer/electric iron/ electric kettle.
	Service and repair of cooking range / geyser/ mixer/grinder / food processor
	Service and repair of induction heater/ fan/ blower/ cooler.
	Service and repair of semi-automatic washing machine.
	Service and repair of refrigerator.
	Explain installation and repair of pump set and submersible pump.
	Carry out repair of electrical circuit of window and split AC.
22. Perform winding of small transformers and motors viz., ceiling fan, table fan, mixer/grinder, submersible pump, etc. (NOS: PSS/N4402)	Perform winding of single-phase transformer.
	Perform winding of ceiling fan / table fan motor.
	Carry out maintenance, service and repair of single-phase AC motors; mixer/grinder, table fan pumps etc.
	Carry out maintenance and servicing of universal motor.
	Carry out winding of submersible pump.
	Carry out winding of 3-φ AC motor.
23. Carry out Estimation & costing for different wiring systems and ready to adopt structured / smart wiring concept for automation and IoT applications. (NOS: N/A)	Perform estimation and costing for different types/scheme of wiring for labour, materials and accessories for a given wiring layout.
24. Read and apply engineering drawing for different application in the field of work.	Read & interpret the information on drawings and apply in executing practical work.
	Read & analyze the specification to ascertain the material requirement, tools and assembly/maintenance parameters.

	Encounter drawings with missing/unspecified key information and make own calculations to fill in missing dimension/parameters to carry out the work.
25. Demonstrate basic mathematical concept and principles to perform practical operations. Understand and explain basic science in the field of study.	Solve different mathematical problems.
	Explain concept of basic science related to the field of study

EMPLOYABILITY SKILLS (CORE SKILL)	
LEARNING OUTCOME	**ASSESSMENT CRITERIA**
1. Introduction to Employability Skills	Outline the importance of Employability Skills for the current job market and future of work.
	List different learning and employability related GOI and private portals and their usage.
	Research and prepare a note on different industries and the available opportunities.

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2. Constitutional values - Citizenship	Explain the essential civic rights and duties required to be followed to become a responsible citizen.
	Discuss the role of personal values and ethics in personal and social development.
	Identify and practice different environmentally sustainable practices.
3 Becoming a Professional in the 21st Century	Discuss relevant 21st century skills required for employment
	Highlight the importance of practicing 21st century skills like Self-Awareness, Behavior Skills, Positive attitude, self - motivation, problem solving, and time management skills in personal or professional life
	Create a pathway for adopting a learning mindset for personal and professional development
4. Basic English Skills	Use appropriate grammar and sentences while interacting with others
	Read English text with appropriate articulation
	Role play a situation on how to talk appropriately to a customer in English, over the phone or in person
	Write a short note/paragraph / letter/e -mail using correct English
5. Career Development & Goal Setting	Research and identify trends and different skills required to match the current market requirement for a job
	Create a career development plan with well-defined short- and long-term goals
6. Communication Skills	Demonstrate how to communicate effectively using verbal and nonverbal communication etiquette
	Write a short note/paragraph on a familiar topic
	Explain the importance of communication etiquette including active listening for effective communication
	Role play a situation on how to work collaboratively with others in a team
7. Diversity and Inclusion	Exhibit how to behave, communicate and conduct oneself appropriately with all genders and PwD
	Discuss the POSH Act and its significance
8. Financial and Legal Literacy	Discuss various financial institutions, products, and services

	Demonstrate how to carry out offline and online financial transactions, safely and securely and check passbook/statement
	Explain the common components of salary such as Basic, PF, Allowances (HRA, TA, DA, etc.), tax deductions
	Calculate income and expenditure for budgeting
	Discuss the legal rights, laws, and aids
9. Essential Digital Skills	Describe the role of digital technology in day-to-day life and the workplace
	Demonstrate how to operate digital devices and use the associated applications and features, safely and securely
	Demonstrate how to connect devices securely to internet using different means
	Follow the dos and don'ts of cyber security to protect against cyber crimes
	Discuss the significance of displaying responsible online behaviour while using various social media platforms
	Create an e-mail id and follow e- mail etiquette to exchange e -mails
	Show how to create documents, spreadsheets and presentations using appropriate applications
10. Entrepreneurship	Describe the types of entrepreneurship and enterprises
	Discuss the process of identifying opportunities for potential business and relevant regulatory and statutory requirements
	Describe the 4Ps of Marketing-Product, Price, Place and Promotion and apply them as per requirement
	Create a sample business plan, for the selected business opportunity
	Discuss various sources of funding and identify associated financial and legal risks with its mitigation plan
11. Customer Service	Describe different types of customers
	Role play a situation on how to identify customer needs and respond to them in a professional manner
	Explain various tools used to collect customer feedback
	Draft a professional Curriculum Vitae (CV)

12. Getting ready for apprenticeship & Jobs	Use various offline and online job search sources such as employment exchanges, recruitment agencies, and job portals respectively
	Demonstrate how to apply to identified job openings using offline /online methods as per requirement
	Discuss the significance of maintaining hygiene and dressing appropriately
	Discuss how to prepare for an interview
	Role play a mock interview
	List the steps for searching and registering for apprenticeship opportunities

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SECTION 2

25. EVIDENCE OF LEVEL

OPTION A

Title/Name of qualification/component: WIREMAN			**Level: 4**
NSQF Domain	**Outcomes of the Qualification/Component**	**How the outcomes relates to the NSQF level descriptors**	**NSQF Level**
Process	**Work in familiar, predictable, routine, situation of clear choice** ● Make good quality electrical wire joints for single and multi strand conductors suitable for applications with soldering following safety precautions. ● Plan, draw, estimate material, wire up and test different type of domestic wiring circuits as per Indian Electricity rules and taking care of quality. Construction and working of MCB & ELCB. Test a domestic wiring installation using Megger. ● Make choices to carry out basic jobs of marking out the components for filing, drilling, and riveting, fitting and assembled using different components independently.	The learner requires to work in familiar and predictable work for example 'Plan, draw, estimate material, wire up and test different type of domestic wiring circuits as per Indian Electricity rules and taking care of quality. Construction and working of MCB & ELCB. Test a domestic wiring installation using Megger'. One needs to perform routine set of activities in a situation of clear choice. The learner requires to 'Make choices to carry out basic jobs of marking out the components for filing, drilling, and riveting, fitting and assembled using different components independently' and 'Plan and install Pipe & Plate earthing'. In all these learning outcomes the learner has to apply one's knowledge to meet the client's requirement in familiar, predictable and routine situation with clear choice of procedures.	4

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Title/Name of qualification/component: WIREMAN			**Level: 4**
NSQF Domain	**Outcomes of the Qualification/Component**	**How the outcomes relates to the NSQF level descriptors**	**NSQF Level**
	● Plan and install Pipe & Plate earthing. Measure earth resistance by earth tester.	Hence NSQF Level is 4 for this descriptor.	
Professional knowledge	**Factual knowledge of field of knowledge or study** ● Introduction and explanation of electrical wiring systems, cleat wiring, casing & Capping, CTS, Conduit and concealed etc. ● Branching of circuits with respect to loads such as lighting and power. ● Conduit pipe wiring materials and accessories, types and sizes of conduit. ● Construction, working & applications of – Incandescent lamp, Fluorescent tube, CFL, Neon sign, Halogen, Mercury vapour and types, sodium vapour etc. ● Wiring in commercial building- their special precautions as per I.E. rules. ● General idea of fixing meter boards & taking service connection.	The learner requires demonstrating factual knowledge of work or study. He applies the processes and general concepts, in the field of work or study related to Wireman Trade like Electrical wiring, branching of circuits, conduit pipe wiring materials etc. Hence NSQF Level is 4 for this descriptor.	4

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Title/Name of qualification/component: WIREMAN			**Level: 4**
NSQF Domain	**Outcomes of the Qualification/Component**	**How the outcomes relates to the NSQF level descriptors**	**NSQF Level**
Professional skill	● Plan, draw, estimate material, wire up and test different type of industrial wiring circuits as per Indian Electricity rules and taking care of quality. ● Construct and test Half–wave, full-wave, and bridge rectifiers with filter & without filter. Troubleshoot and service of DC regulated power supply. ● Select and perform electrical/ electronic measurements with appropriate instrument.	As per the learning outcomes indicated in the adjacent cell the learner recalls and demonstrates practical skill such as 'Plan, draw, estimate material, wire up and test different type of industrial wiring circuits as per Indian Electricity rules and taking care of quality' and 'Select and perform electrical/ electronic measurements with appropriate instrument' applying basic methods, tools, materials and information. Hence NSQF Level is 4 for this descriptor.	4
Core skill	**Basic Mathematical and Algebraic principles** ● Demonstrate basic mathematical concept and principles to perform practical operations. ● Understand and explain basic science in the field of study including simple machine. ● Read and apply engineering drawing for different application in the field of work. **Basic understanding of social political and natural environment** ● Explain energy conservation, global warming and pollution and contribute in day to	The learning outcomes for example 'Demonstrate basic mathematical concept and principles to perform practical operations' and 'Understand and explain basic science in the field of study including simple machine ' display the learning outcomes where the learner needs to display basic mathematical and algebraic principles; understanding of social, political; and some skill of collecting and organising information, communication. Hence NSQF Level is 4 for this descriptor.	4

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Title/Name of qualification/component: WIREMAN			**Level: 4**
NSQF Domain	**Outcomes of the Qualification/Component**	**How the outcomes relates to the NSQF level descriptors**	**NSQF Level**
	day work by optimally using available resources. ● Explain personnel finance, entrepreneurship and manage/organize related task in day to day work for personal & societal growth. **Language to communicate written or oral, with required clarity** ● Interpret & use formal and technical communication. ● List and interpret various acts of labour welfare legislation.		
Responsibility	● Interpret the types, constructional features, working principles of transformer (single & three phase) Connect and test Transformer. ● Prepare single line diagram and layout plan of electrical transmission & distribution systems and power plants with knowledge of principle applied. Make and test power connection to substation equipments with care and safety.	The role of Wireman is independently responsible to perform the work as per specifications and their own analysis of what needs to be done based on their understanding of electrical processes, principles and standards. Learning outcomes like “Select, assemble, test and wire-up control panel”, “Plan, estimate and costing of different types of wiring system as per Indian Electricity rule” etc. reveal the same.	4

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Title/Name of qualification/component: WIREMAN			**Level: 4**
NSQF Domain	**Outcomes of the Qualification/Component**	**How the outcomes relates to the NSQF level descriptors**	**NSQF Level**
	● Select, assemble, test and wire-up control panel. ● Plan, estimate and costing of different types of wiring system as per Indian Electricity rule.	Hence NSQF Level is 4 for this descriptor.	

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SECTION 3
EVIDENCE OF NEED

26 **What evidence is there that the qualification is needed? What is the estimated uptake of this qualification and what is the basis of this estimate?**

Basis	In case of other Awarding Bodies (Institutes under Central Ministries and states departments)
Need of the qualification	Power Sector has a significant presence of organized as well as unorganized skilled manpower requirement. This sector is poised to grow exponentially in the years to come and is highly labour intensive and there are many emerging trends in this sector. Hence the qualification has been designed keeping in view to cater to the ever-increasing demand of skilled manpower in consultation with stakeholders.
Industry Relevance	The job role defined for the qualification is as per the National Classification of Occupations 2015 which is developed by Employment Directorate under the ministry of Labour and Employment in collaboration with different industry partners and as per ILO guidelines. Moreover, the training is imparted in ITIs/NSTIs/MSTIs/BTC/ BTPs/ Industries / Establishments etc. where such requirement is available. This justifies the qualification is very much relevant for industry.
Usage of the qualification	The Proposed qualification will create skilled Technician for various establishments in different Sectors.
Estimated uptake	The present seating capacity is 62118.

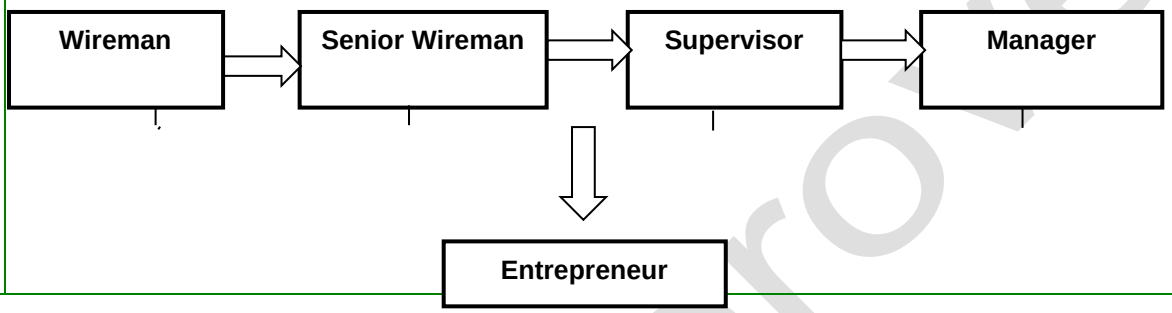
27 **Recommendation from the concerned Line Ministry of the Government/Regulatory Body. To be supported by documentary evidences.**

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	The qualification originally designed for Craftsman Training Scheme is in existence for many years and approved by DGT (Regulatory Body) under Ministry of Skill Development and Entrepreneurship, Govt. of India.
28	**What steps were taken to ensure that the qualification(s) does (do) not duplicate already existing or planned qualifications in the NSQF? Give justification for presenting a duplicate qualification** The qualification is originally designed and approved by DGT for the Craftsman Training Scheme and is in existence for many years. No such duplicate qualification of same duration and competencies exists.
29	**What arrangements are in place to monitor and review the qualification(s)? What data will be used and at what point will the qualification(s) be revised or updated? Specify the review process here** ● The research wing of CSTARI & DGT reviews and updates the qualification, in consultation with industries and other stakeholders, on a regular basis by conducting trade committee meetings. ● DGT will monitor any duplicity by comparing existing qualifications with upcoming ones in the National Qualifications Register (NQR) and relevant sectors.

SECTION 4
EVIDENCE OF PROGRESSION

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30	**What steps have been taken in the design of this or other qualifications to ensure that there is a clear path to other qualifications in this sector?** ***Show the career map here to reflect the clear progression*** On completion of the training the trainee will have an opportunity to move in vertical/horizontal pathways to promote to higher designations. The learner can further undergo other specialised courses to excel in the relevant field. 

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